

Lecture 5

Data Exploration and Data Cleaning

Expanded concept and example edition

Data Analytics for Personalized Genomics and Precision Medicine

Lecturer Prof. Yu Li Li Ruixuan

Expanded companion to Lecture 5 Working Notes (BMEG3105_Lec5_script). The same core concepts and classroom examples are developed step by step. Supplementary examples add explanation and are distinguished from the lecture material.

5.1 From sequence reads to a data matrix

The lecture connects the sequence analysis introduced previously to the numerical data used in genomics. In the simplified RNA-seq workflow, reads are mapped to a reference genome, assigned to genes, and counted. Repeating this process for multiple samples produces a gene-by-sample count matrix. The next task is to check whether that matrix is suitable for analysis.

5.1.1 Genome assembly

Assembly reconstructs a longer sequence from overlapping reads. The lecture contrasts short reads of roughly 200 bases with a human genome of about 3 billion bases. The jigsaw-puzzle analogy is useful: each read provides local information, and consistent overlaps connect those local pieces into a longer sequence. [1, pp. 8–12]

Short reads overlap by two bases in this toy construction



Figure 5.1 Reconstructed overlap example from the lecture. The supplied 3-base reads form the 14-base sequence TAATGCCATGGATG; this illustrates one consistent assembly.

An overlap does not always determine a unique placement. Repeated ATG reads can fit at several positions, sequencing errors can create conflicting bases, and mutations can complicate the comparison. Finding and checking overlaps across a large collection of reads also creates a computational challenge.

Classroom example 5.1 Repeated sequence. If every read inside a long run of A bases is only AAAA, overlap information alone cannot determine the run length. Many different lengths are compatible with the same short-read sequence. A sufficiently long read that spans the repetitive region can supply the missing information. The limitation is in the available evidence, not merely the speed of the algorithm. [7, Section 5.1]

5.1.2 Mapping reads and counting expression

Mapping starts with an existing reference sequence and asks where each read best fits. The classroom method slides a read along the genome and counts mismatches at each position. This simplified comparison does not allow gaps; more general alignment methods can use dynamic programming. The lecture introduces the principle rather than a production mapping algorithm. [1, pp. 15–23]

Example 5.2 Mapping CCA. For TAATGCCATGGATG, the mismatch counts at starting positions 1 through 12 are 2, 3, 3, 3, 2, 0, 2, 3, 3, 2, 3, 3. The exact match starts at position 6, using one-based positions.

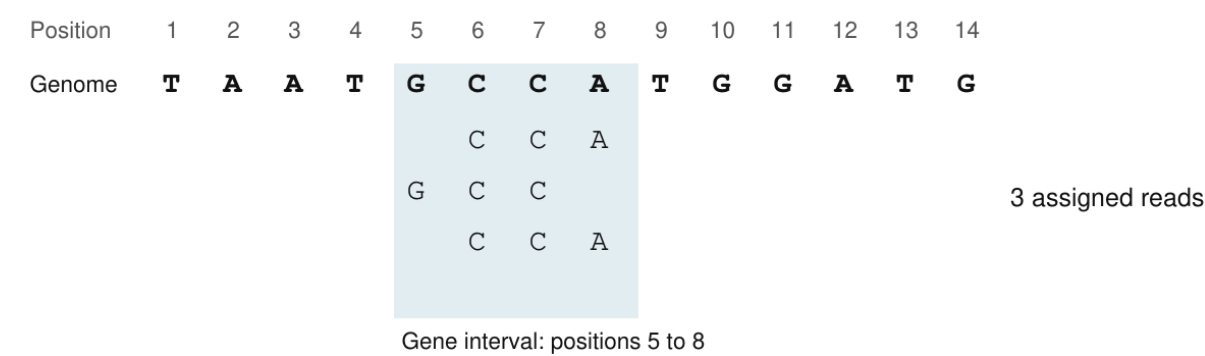


Figure 5.2 The highlighted GCCA gene interval receives GCC and two CCA reads, giving a count of 3 under the illustrated assignment rule. Redrawn from [1, p. 20].

The additional read ATG matches positions 3–5, 8–10, and 12–14. Two placements partly overlap the highlighted gene. The teacher emphasized that ambiguous placement and partial overlap require an explicit rule: a method may exclude the read or assign it to the gene. Accordingly, a count of 3 or 4 depends on that rule. State the assumption when solving the exercise. [7, Section 5.1]

5.1.3 What the matrix represents

For a general data matrix X with n rows and m columns, each row represents an object and each column an attribute. In the height-and-weight example, four people and two measured attributes give a 4×2 matrix; the person identifier is a label, not a measured feature. Reordering entire rows or columns preserves the information only when the associated labels move with them. Shuffling individual cells does not.

The displayed gene-expression matrix uses a different orientation: rows are genes and columns are samples. Each cell records the value for one gene in one sample. Always check the axis labels before normalizing, comparing, or interpreting the matrix. Raw read counts are an input to expression analysis, rather than an automatically comparable expression scale.

Stage	Input	Output
Assembly	Overlapping sequence reads	A reconstructed sequence
Mapping	Reads and a reference	Candidate read positions
Counting	Assigned reads and gene intervals	A gene-by-sample count matrix

Project connection. The suggested project is to obtain a reference and reads, use existing software for mapping and counting, and explain each stage’s input, output, and purpose. Writing an assembler or mapper from scratch is not required. [7, Section 5.1]

5.2 Understanding data quality problems

Classroom basis: [1, pp. 27–32; 7, Section 5.2]. The worked numerical illustrations on this page are supplementary.

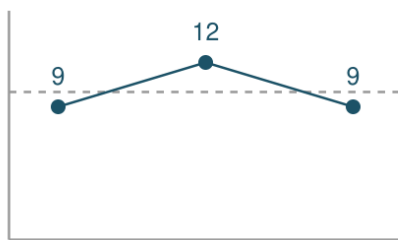
Cleaning is a decision about what an observation means. A fluctuating signal, an exceptional individual, a missing measurement, and a repeated record are not the same problem. The lecture asks us to identify the problem, find evidence for it, and choose a treatment that fits the data.

5.2.1 Noise and the reason for smoothing

Noise modifies the underlying values. The teacher uses a noisy sine wave and explains that averaging nearby observations can reduce random fluctuations. The key assumption is that nearby positions contain related signal. Averaging unrelated rows in a people table would not have the same justification.

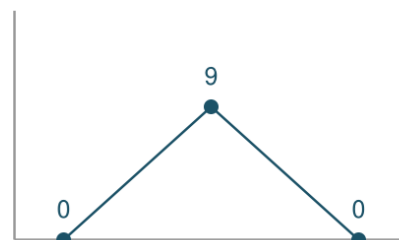
Supplementary example S1. Suppose a locally constant signal should be 10, but three neighboring readings are 9, 12, and 9. Replacing the center by their average gives $(9 + 12 + 9)/3 = 10$. Here the neighboring errors cancel, recovering the assumed signal.

Locally constant signal



Center after averaging = 10

A real short pulse



Center after averaging = 3

Figure 5.3 The same three-point averaging rule can recover a constant signal or flatten a genuine pulse. Constructed values illustrate the assumption behind smoothing.

Now suppose the observations 0, 9, 0 describe a real short pulse. The same average is 3, so smoothing suppresses the peak from 9 to 3. A smoother curve is not automatically a more faithful measurement. Decide what changes are noise before choosing a smoothing rule.

5.2.2 Outliers need an explanation

An outlier differs considerably from most other objects. The teacher contrasts erroneous entries with a student who genuinely scores 95 when most classmates score around 60. The exceptional student is unusual, but the score need not be incorrect. [7, Section 5.2]

Observation	Question to ask	Possible response
A recorded height of 11 m	Unit or entry error?	Check the source; correct only with evidence.
A score of 95 among scores near 60	Is this a valid exceptional result?	Retain if the measurement is valid.
Small fluctuations around a smooth signal	Are local deviations random noise?	Consider smoothing with a justified neighborhood.

The first two situations both look unusual, but they support different decisions. A numerical rule can flag a record for inspection; context determines whether correction, exclusion, or retention is appropriate.

5.2.3 Missing values and duplicate records

A missing value may mean that information was not collected, a person declined to provide it, or the attribute does not apply. The lecture describes removing the object, estimating the missing value, ignoring the missing entry where a method permits it, or considering plausible alternatives weighted by their probabilities.

Classroom example. P2 has height 1.64 m but no recorded weight. In the original four-person table, P3 at 1.70 m has the closest observed height and weighs 63 kg. The teacher uses this similarity to motivate 63 kg as an estimate. It is a prediction based on an assumption, not a newly observed weight. [7, Section 5.2]

Supplementary example S2 Compare the consequences. The observed weights are 75, missing, 63, and 78 kg. Consider three ways to proceed:

Treatment	Calculation	Interpretation
Exclude the missing value	$(75 + 63 + 78)/3 = 72$	Mean of the three observed weights only.
Impute the nearest-height value	$(75 + 63 + 63 + 78)/4 = 69.75$	Mean includes the estimated value of 63.
Replace the missing value by zero	$(75 + 0 + 63 + 78)/4 = 54$	Treats missingness as a real zero-weight observation.

The third calculation demonstrates why “missing” and zero are not interchangeable. The first two are different analyses with different assumptions. A complete-looking table does not remove the uncertainty introduced by estimation.

Why duplicates change the result

When databases overlap, a person may occur more than once. This silently increases that person’s weight in the calculation. The teacher’s extreme example puts 94 copies of P1 among 100 records, pulling the mean toward P1’s height of 1.79 m. [7, Section 5.2]

Supplementary example S3. For four distinct people with weights 54, 63, 75, and 78 kg, the mean is $270/4 = 67.5$ kg. Accidentally adding a second copy of the 75 kg record gives $(270 + 75)/5 = 69$ kg. Nobody gained weight; the sample representation changed.

This also explains why deduplication must use record identity. Two different people may both weigh 75 kg. Deleting one solely because the weights match would remove a legitimate observation. In the classroom exercise, the repeated P1 identifier and matching fields establish the duplicate.

Identify the record → Check why it is repeated → Keep one copy if duplicated

Cleaning decisions should be traceable: distinguish measured values from imputed values, retain a reason for exclusions, and state how repeated observations were identified. This makes later summaries interpretable.

5.3 Scaling and representing attributes

Classroom basis: [1, pp. 33–36; 7, Section 5.3].

5.3.1 Why numerical scale matters

For P1 = (1.79 m, 75 kg) and P2 = (1.64 m, 54 kg), the numerical height difference is 0.15 while the weight difference is 21. In the sum of squared coordinate differences, height contributes 0.0225 and weight contributes 441. The distance is therefore almost entirely determined by the weight coordinate.

$$d(P1, P2) = \sqrt{0.15^2 + 21^2} \approx 21.0005$$

Supplementary example S4 Change only the unit. Writing the same heights in centimeters makes their difference 15 instead of 0.15. Its squared contribution becomes 225 instead of 0.0225. The people are unchanged, but height now contributes $225/(225 + 441) \approx 33.8\%$ of the squared distance. Raw mixed-unit distances therefore reflect the representation as well as the differences between people.

5.3.2 Min-max normalization

For a nonconstant attribute, subtract the minimum and divide by the range. Compute these values separately for each attribute. The observed minimum becomes 0 and the observed maximum becomes 1.

$$v' = \frac{v - v_{min}}{v_{max} - v_{min}}$$

Classroom calculation. The height range is 1.64–1.88 m and the weight range is 54–78 kg. P1's scaled height is $(1.79 - 1.64)/0.24 = 0.625$; its scaled weight is $(75 - 54)/24 = 0.875$.

Person	Height (m)	Weight (kg)	Scaled height	Scaled weight
P1	1.79	75	0.625	0.875
P2	1.64	54	0	0
P3	1.70	63	0.250	0.375
P4	1.88	78	1	1

For P1 versus P2, the squared terms are now $0.625^2 = 0.390625$ and $0.875^2 = 0.765625$. Both attributes contribute visibly, and the distance is about 1.0753. Scaling makes their numerical ranges comparable; it does not guarantee equal contributions to every pairwise distance.

Check the limits. If every value in a column is identical, the range is zero and the formula needs a separate convention. Scaling also does not remove a bad measurement: an erroneous extreme still determines the range. The complete exercise in Section 5.4 shows the consequence. [3, 5]

5.3.3 Z-scores and the normalization direction

Z-score standardization expresses distance from an attribute's mean in standard-deviation units. A positive z-score is above the mean; a negative one is below it. State whether the standard deviation uses a sample or population convention.

$$z = \frac{v - \mu}{\sigma}$$

Supplementary example S5. For values 2, 4, 6, the mean is 4. With the population convention, the variance is $[(2 - 4)^2 + (4 - 4)^2 + (6 - 4)^2]/3 = 8/3$. The standard deviation is about 1.633, giving z-scores -1.225, 0, and 1.225. The standardized values have mean 0 and population variance 1.

Clarification: the slide associates z-scores with Gaussian data. The arithmetic transformation does not require a Gaussian distribution and does not turn arbitrary data into a Gaussian distribution. [3]

For the people table, height is scaled across people and weight is scaled separately. Mixing height and weight within one person would combine different attributes. A gene-by-sample matrix requires an additional decision: normalize across genes within a sample, or compare a gene across samples? The teacher emphasizes explaining the purpose and direction. [7, Section 5.3]

Supplementary example S6 Technical depth versus composition. Suppose two samples have the following constructed counts:

Gene	Sample A	Sample B	A / column total	B / column total
G1	100	200	0.25	0.25
G2	300	600	0.75	0.75
Total	400	800	1	1

Every raw count in B is twice its value in A, but dividing by each sample's total gives identical proportions. This illustrates the teacher's point that more reads can create larger counts without a change in relative composition. It does not prove the cause of a real sample difference, and total-count division is not a complete RNA-seq normalization or batch-correction procedure.

5.3.4 One-hot encoding

A categorical attribute is valid data, but its labels are not numerical measurements. The classroom color example creates one indicator per category: Red = (1, 0, 0), Blue = (0, 1, 0), and Green = (0, 0, 1), in the fixed column order Red, Blue, Green.

Supplementary example S7 Why not use 1, 2, 3? Coding Red = 1, Blue = 2, Green = 3 makes Red-Green numerically twice as far apart as Red-Blue, although the categories have no stated order. With one-hot vectors, every pair of different colors has Euclidean distance $\sqrt{2}$. This removes the arbitrary ordering; it does not claim that all colors are equally similar perceptually.

5.4 Putting the cleaning steps together

Source: [1, p. 38; 7, Section 5.4]. Final scaled values are calculated here to complete the classroom example.

Merge the two input tables while preserving the meaning of each person's record. A dash denotes a missing weight, not zero.

Source	Person	Height (m)	Weight (kg)	Color
A	P1	1.79	75	Red
A	P2	1.64	—	Blue
A	P3	1.69	63	Green
A	P4	1.85	78	Red
B	P1	1.79	75	Red
B	P6	1.64	54	Green
B	P7	1.70	1000	Blue
B	P8	1.88	80	Green

First, flag P7's 1000 kg as an implausible entry and treat that weight as missing rather than guessing the intended number. Following the classroom nearest-height rule, fill P2's weight with P6's 54 kg, and P7's weight with P3's 63 kg. Then keep one copy of P1, encode color, and scale height and weight separately.

After cleaning, the height range is 1.64–1.88 m and the weight range is 54–80 kg. The resulting table has seven people and five numerical features; the person identifier is retained as a label.

Person	Height scaled	Weight scaled	Red	Blue	Green
P1	0.6250	0.8077	1	0	0
P2	0.0000	0.0000	0	1	0
P3	0.2083	0.3462	0	0	1
P4	0.8750	0.9231	1	0	0
P6	0.0000	0.0000	0	0	1
P7	0.2500	0.3462	0	1	0
P8	1.0000	1.0000	0	0	1

Why order matters. If min-max scaling is performed while 1000 remains in the weight column, a valid 75 kg becomes $(75 - 54)/(1000 - 54) \approx 0.0222$. After treating the error, it becomes $(75 - 54)/(80 - 54) \approx 0.8077$. The erroneous maximum compresses the valid weights into a narrow interval and distorts later comparisons. [7, Section 5.4]

There is no universally optimal order for every dataset. Here the important dependency is to resolve the implausible value before estimating the scaling parameters. The two imputed weights are estimates and should remain identifiable as such in the analysis record.

5.5 Reading summary statistics

Classroom basis: [1, pp. 41–46; 7, Section 5.5]. The added calculations make the definitions explicit.

Location describes where the data lie; spread describes how widely they vary. Frequency describes how often a value or category occurs. The lecture uses GPA as an intuition: an individual value is easier to interpret when we also know the distribution around it.

5.5.1 Mean and median

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The mean uses every value. To obtain the median, first sort the data; take the middle value for an odd number of observations or average the two middle values for an even number. A trimmed mean instead discards a chosen fraction from each tail before averaging.

Worked extension of the classroom table. For weights 75, 54, 63, and 78 kg, the total is 270 and the mean is $270/4 = 67.5$ kg. Sorting gives 54, 63, 75, 78, so the median is $(63 + 75)/2 = 69$ kg. Averaging the middle two entries before sorting would be a different and incorrect procedure.

5.5.2 Range and variance

$$Range = x_{max} - x_{min} \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Weight	Deviation from 67.5	Squared deviation
54	-13.5	182.25
63	-4.5	20.25
75	7.5	56.25
78	10.5	110.25
Total	0	369

The range is $78 - 54 = 24$ kg. Sample variance is $369/(4 - 1) = 123 \text{ kg}^2$, and sample standard deviation is $\sqrt{123} \approx 11.09$ kg. Squaring prevents positive and negative deviations from cancelling; taking the square root restores the original unit.

With a population denominator, variance is $369/4 = 92.25 \text{ kg}^2$ instead. The teacher accepts a clearly stated convention. The $n - 1$ denominator is used for the usual unbiased estimate of a population variance from independent observations; it is not a universal replacement for n . [7, Section 5.5]

Supplementary example S8 One extreme value. Changing $[2, 3, 4, 5, 6]$ to $[2, 3, 4, 5, 60]$ moves the mean from 4 to 14.8 and the sample variance from 2.5 to 639.7. The median stays at 4. This is why the mean alone can be misleading when an extreme value is present.

5.5.3 Robust spread and percentiles

Median absolute deviation (MAD) summarizes the typical absolute deviation from the median. Absolute values avoid cancellation; the outer median reduces the influence of a very large deviation. [4]

$$MAD = \text{median}(|x_i - \text{median}(x)|)$$

For weights 54, 63, 75, 78, the median is 69. The absolute deviations are 15, 6, 6, 9; sorting gives 6, 6, 9, 15. Thus $MAD = (6 + 9)/2 = 7.5$ kg. This is the unscaled MAD. For both datasets in example S8, MAD is 1 even though the largest observation changes dramatically.

Formula clarification: the slide centers deviations on the mean under the name “median absolute deviation.” The standard definition used here centers them on the median. This supplies the definition for the spread measure listed in the working notes. [1, p. 44; 4; 7, Section 5.5]

The percentile question from class

For values 1, 2, ..., 10, the teacher asks which observed value has 30% of the observations strictly below it. The answer under that stated rule is 4: the three values 1, 2, 3 are below 4. If the rule is changed to “at or below,” the classroom answer becomes 3. [7, Section 5.5]

Rule	Candidate	Number counted	Proportion
Strictly below	3	2	20%
Strictly below	4	3	30%
At or below	3	3	30%

This example teaches definition-checking. In a finite dataset, quantiles also require a rank or interpolation convention; ties may make an exact proportion impossible. Do not treat the classroom counting rule as the only software definition of a percentile.

Supplementary example S9 Interquartile range. Using the median-of-halves convention for 1 through 10, the lower half is 1–5 and the upper half is 6–10. Their medians are $Q1 = 3$ and $Q3 = 8$, so $IQR = 8 - 3 = 5$. The IQR describes the middle portion of the distribution. This stated convention is separate from the strict-count classroom question above.

5.5.4 Frequency and mode

Relative frequency is count divided by the total number of observations; the mode is the most frequent value. In the cleaned seven-person table, Green appears three times and Red and Blue twice each. Their relative frequencies are $3/7$, $2/7$, and $2/7$, and Green is the mode. A frequency is meaningful for a category even when an arithmetic mean of the category names is not.

5.6 Using visualization to understand the data

Classroom basis: [1, pp. 48–51; 7, Section 5.6]. Histogram arithmetic below is supplementary.

The sea-surface-temperature example turns records of (longitude, latitude, temperature) into a map. Location encodes the first two values and color encodes the third. Broad patterns become visible without reading thousands of triplets. The point is to choose visual encodings that match the meaning of the data.

5.6.1 Constructing a histogram

A histogram groups one numerical variable into intervals and counts observations in each interval. It shows a distribution rather than the identity of each observation. Bin choices affect what structure is visible. [6]

Supplementary example S10. Use the constructed values 1, 1, 2, 2, 3, 3, 4, 4, 8, 9. With bins $[0, 2)$, $[2, 4)$, $[4, 6)$, $[6, 8)$, and $[8, 10]$, the counts are 2, 4, 2, 0, and 2. All but the last interval exclude their right endpoint. For example, both observations equal to 2 go into $[2, 4)$, not $[0, 2)$.

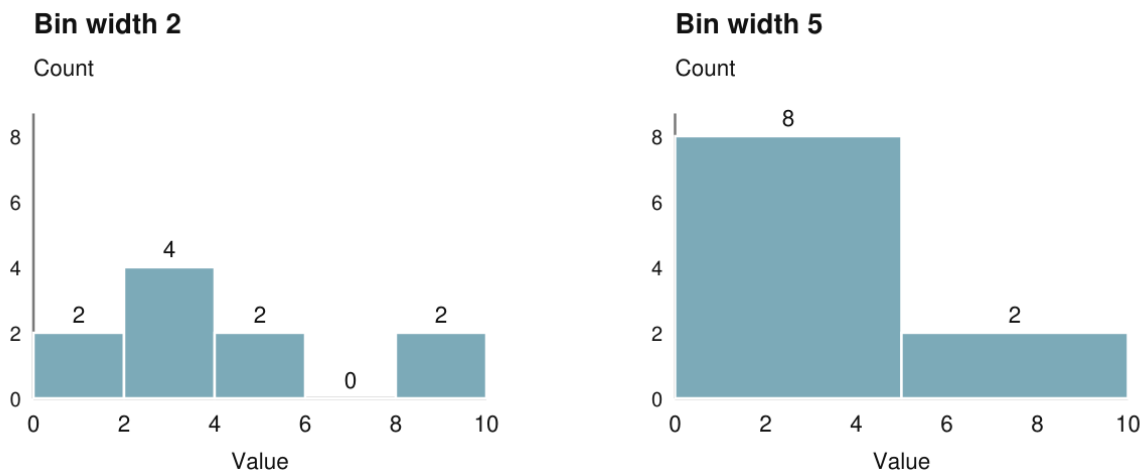


Figure 5.4 Two histograms of the same ten constructed observations. Both count every observation once, but the wider bins conceal the empty interval from 6 to 8.

With only the bins $[0, 5)$ and $[5, 10]$, the counts become 8 and 2. The first plot reveals a gap; the second summarizes the lower-versus-higher split. Neither binning changes the data. A pattern that appears only under one arbitrary binning deserves further checking.

5.6.2 Connect the plot to a cleaning decision

The teacher briefly uses a height of 11 m as an example of an implausible value that visualization can reveal. Finding an isolated observation is a diagnostic step: return to the units and source record before deciding what to change. A graph should lead to a checkable explanation, not automatic deletion.

Plot or summarize → Investigate unusual values → Clean if justified → Check again

Consistent with the working notes, this section covers introductory visualization. Box plots, clustering, and later similarity and distance topics are outside its scope. Euclidean distance is used here to explain the motivation for normalization. [7, Sections 5.6–5.7]

5.7 Connecting the concepts

The workflow starts with biological measurements and ends with interpretable summaries. Every intermediate choice can change the final answer: ambiguous reads affect counts; duplicates affect weighting; imputation affects the data distribution; scaling affects numerical comparisons; percentile and variance conventions affect reported statistics.

Reads → Mapping and counting → Quality checks → Representation → Exploration

What to explain in a worked answer. Name the problem, state the assumption, show the calculation, and interpret the result. For example, “P7’s 1000 kg is treated as missing; the nearest-height record supplies an estimate; scaling then uses the corrected range.” This explains more than a list of cleaning operations.

Sources and example provenance

Slide references use physical PDF pages. Reference [7] identifies the companion working notes and their section numbers. Supplementary examples S1–S10 use constructed datasets and calculations to explain the concepts; they are not empirical study results or examples attributed to the lecturer.

[1] Li, Y. Data exploration and data cleaning. Supplied BMEG3105 Lecture 5 slides, lec5-Exploration and cleaning.pdf. Relevant material: PDF pp. 3–51.

[2] Li, Y. From sequence to gene expression matrix: Assembly and mapping. Supplied lec4-Assembly and mapping.pdf, used for context.

[7] Li, Ruixuan. Lecture 5 Working Notes. BMEG3105_Lec5_script.docx. Companion notes used to align the topic sequence and classroom examples.

[3] scikit-learn developers. Preprocessing data. <https://scikit-learn.org/stable/modules/preprocessing.html>. Definition and interpretation of scaling.

[4] NIST. Median Absolute Deviation. <https://itl.nist.gov/div898/software/dataplot/refman2/auxillar/mad.htm>. Standard MAD definition.

[5] scikit-learn developers. Compare the effect of different scalers on data with outliers. https://scikit-learn.org/stable/auto_examples/preprocessing/plot_all_scaling.html. Sensitivity of standard and min–max scaling to outliers.

[6] NIST/SEMATECH e-Handbook. Histogram. <https://www.itl.nist.gov/div898/handbook/eda/section3/eda33e.htm>. Histogram definition and interpretation.

AI assistance

OpenAI Codex assisted in locating supplementary reference material, developing additional worked examples, expanding contextual explanations and logical connections, and organizing diagrams and document formatting. Its contribution also included drafting and revising explanatory text and checking numerical calculations. The supplied working notes and course slides guide the scope and core examples; supplementary material is identified separately. Citation: OpenAI. (2026). Codex. <https://openai.com/codex/>.

Relationship to the working notes: Sections 5.1–5.7 follow the same conceptual sequence, from sequence reads and data matrices through cleaning, scaling, summary statistics, and visualization. This expanded version adds calculations, explanations, and referenced clarifications around those topics.

Format reference: Lec5-Scribing-Wang Jiuming.pdf, used for presentation only. The original student example contributes no subject content. Course materials remain subject to the instructor’s redistribution rules.